

Module 4: Fuzzy Logic and Expert Systems

Fuzzy Logic

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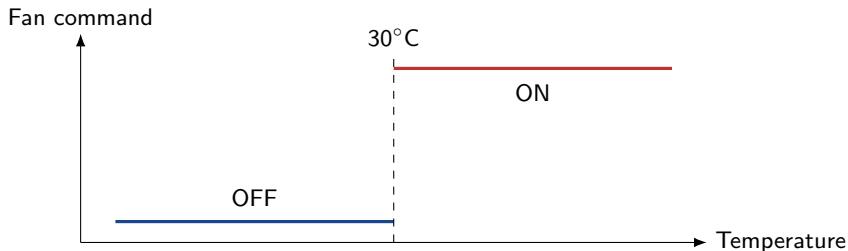
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Why Ordinary Thresholds Become a Problem

Suppose the fan rule is

$$T < 30^{\circ}\text{C} \Rightarrow \text{fan OFF}, \quad T \geq 30^{\circ}\text{C} \Rightarrow \text{fan ON}.$$



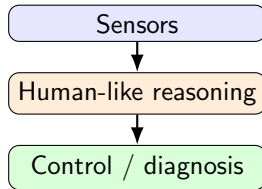
At 29.9°C the fan is OFF; at 30.0°C it suddenly becomes fully ON. Sensor noise can therefore cause rapid switching.

Many physical situations change gradually, not abruptly.

The Story Begins: Decisions Are Not Always Binary

An engineer must automate three decisions:

- adjust a cooling fan using temperature and humidity;
- reduce cobot speed when a person approaches;
- diagnose a motor using vibration and temperature.



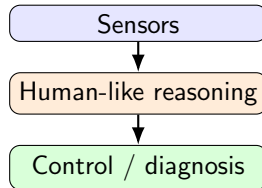
The sensors provide precise numbers, but the technician uses linguistic descriptions:

"The motor is slightly hot, vibration is rather high, and the load is heavy."

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A Simple Human Example

Expression	Degree of liking
I do not like you	0.0
I like you	0.2
I like you as a friend	0.4
I like you	0.7
I like you very much	0.9
I love you	1.0

Represents gradual concepts using values between 0 and 1, instead of restricting every decision to completely false or completely true. Connects numerical measurements with linguistic engineering knowledge.

Instead of asking only

“Is the temperature hot?” $\in \{0, 1\}$,

ask

“To what degree is the temperature hot?” $\in [0, 1]$.

At 28°C

The temperature may belong to *Warm* with degree 0.7 and to *Hot* with degree 0.3 simultaneously.

Important distinction

The value 0.3 is not a 30% probability that the temperature is hot. It is a **degree of membership** in the fuzzy concept *Hot*.

Crisp set

$$\mu_A(x) \in \{0, 1\}.$$

An element is either inside or outside the set.

- $T = 29.9^{\circ}\text{C}$: not hot
- $T = 30.0^{\circ}\text{C}$: hot

Fuzzy set

$$\mu_A(x) \in [0, 1].$$

An element can belong partially to a set.

- $T = 29.9^{\circ}\text{C}$: hot to degree 0.49
- $T = 30.0^{\circ}\text{C}$: hot to degree 0.50

Fuzziness represents gradual boundaries; it does not mean careless or inaccurate reasoning.

For a universe of discourse X , a fuzzy set A is defined by

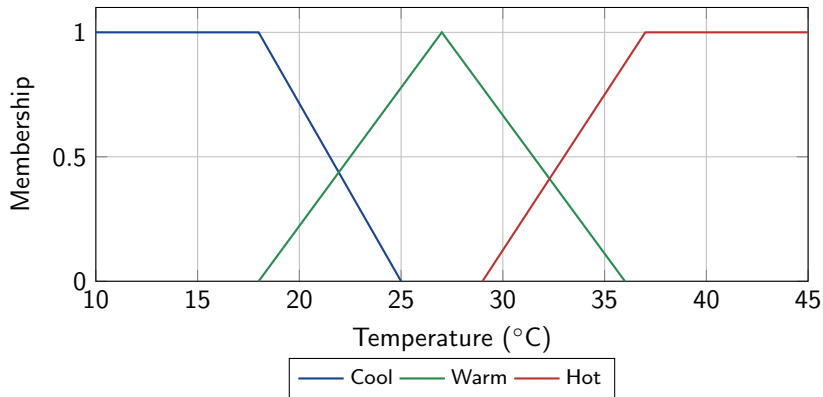
$$A = \{(x, \mu_A(x)) \mid x \in X\}, \quad \mu_A : X \rightarrow [0, 1].$$

- X : all possible values, e.g. 0–50°C.
- $\mu_A(x)$: membership function of the concept A .
- $\mu_A(x) = 0$: no membership.
- $\mu_A(x) = 1$: full membership.
- $0 < \mu_A(x) < 1$: partial membership.

Example

$\mu_{\text{Hot}}(32) = 0.75$ means that 32°C belongs to the fuzzy set *Hot* with degree 0.75.

Membership Functions: Turning Words into Curves



Overlapping curves allow a measurement to activate several linguistic descriptions at the same time.

Common Membership-Function Shapes

Triangular

$$\mu(x; a, b, c) = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right).$$

Trapezoidal

Defined by four points (a, b, c, d) ; useful when a range should have full membership.

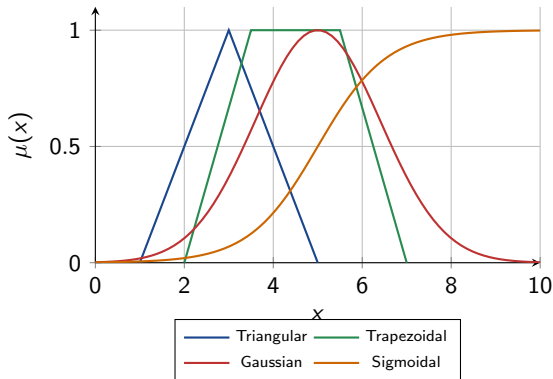
Gaussian

$$\mu(x; c, \sigma) = \exp\left[-\frac{(x-c)^2}{2\sigma^2}\right].$$

Sigmoidal

$$\mu(x; a, c) = \frac{1}{1 + e^{-a(x-c)}}.$$

Useful for open-ended concepts such as *high* or *low*.



Reference: https://www.researchgate.net/publication/33817004_Dimensionality_in_fuzzy_systems/citations

Worked Example: Membership at $T = 32^\circ\text{C}$

Let *Warm* be triangular with $(a, b, c) = (20, 30, 40)$:

$$\mu_{\text{Warm}}(32) = \min\left(\frac{32 - 20}{30 - 20}, \frac{40 - 32}{40 - 30}\right) = \min(1.2, 0.8) = 0.8.$$

Let *Hot* rise linearly from 30°C to 40°C :

$$\mu_{\text{Hot}}(32) = \frac{32 - 30}{40 - 30} = 0.2.$$

Therefore,

$$32^\circ\text{C} = 0.8/\text{Warm} + 0.2/\text{Hot}.$$

A single crisp sensor value can have several valid linguistic interpretations.

- 1 Define the physical operating range.
- 2 Choose meaningful linguistic labels with a domain expert.
- 3 Begin with simple triangular or trapezoidal functions.
- 4 Provide enough overlap for smooth transitions.
- 5 Validate the resulting control or classification behaviour.
- 6 Tune using experiments, optimization, or data-driven learning.

Engineering caution

Membership functions are modelling choices. They must reflect the application and should not be selected only because a curve looks attractive.

For fuzzy sets A and B , commonly used Zadeh operators are

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x)),$$

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x)),$$

$$\mu_{\bar{A}}(x) = 1 - \mu_A(x).$$

Example

If a motor is *Hot* to degree 0.7 and has *High vibration* to degree 0.4, then

$$\mu_{\text{Hot AND HighVibration}} = \min(0.7, 0.4) = 0.4.$$

Operation	Common operator	Alternative
AND	$\min(a, b)$	product ab
OR	$\max(a, b)$	$a + b - ab$
NOT	$1 - a$	application-dependent

For $a = 0.7$ and $b = 0.4$:

$$\min(a, b) = 0.4, \quad ab = 0.28.$$

The selected operator changes the rule firing strength and therefore the final output.

Use one consistent operator family and validate it for the intended system.

A linguistic variable is described by words rather than only by numbers:

$$\mathcal{L} = (x, T(x), X, G, M).$$

- x : variable name, e.g. Temperature.
- $T(x)$: linguistic terms, e.g. {Cool, Warm, Hot}.
- X : numerical universe, e.g. $[0, 50]^{\circ}\text{C}$.
- G : rules for forming terms, e.g. *very hot*, *not hot*.
- M : membership function assigned to each term.

Mechatronics examples

Distance: {Near, Moderate, Far}; vibration: {Low, Medium, High}; cobot speed: {Stop, Slow, Normal}.

If $\mu_A(x)$ describes *Hot*, common hedges may be modelled as

$$\mu_{\text{Very } A}(x) = \mu_A^2(x), \quad \mu_{\text{MoreOrLess } A}(x) = \sqrt{\mu_A(x)},$$

$$\mu_{\text{Not } A}(x) = 1 - \mu_A(x).$$

If $\mu_{\text{Hot}}(x) = 0.7$, then

$$\mu_{\text{VeryHot}} = 0.49, \quad \mu_{\text{MoreOrLessHot}} \approx 0.84.$$

Hedges allow expert language to be represented mathematically.

An experienced technician says:

“If the motor is hot and vibration is high, reduce the load and inspect the bearing.”

The fuzzy rule becomes

IF Temperature is Hot AND Vibration is High THEN Risk is Critical.

- Antecedent: condition after IF.
- Consequent: decision after THEN.
- Rule strength: degree to which the antecedent is satisfied.

Sensor readings give

$$\mu_{\text{Hot}}(T) = 0.7, \quad \mu_{\text{HighVibration}}(V) = 0.4.$$

For the rule

IF T is Hot AND V is High THEN Risk is Critical,

using the minimum AND operator,

$$w = \min(0.7, 0.4) = 0.4.$$

The consequent *Critical* is therefore activated to degree 0.4.

Interpretation

The rule is neither completely inactive nor completely active; it contributes in proportion to its firing strength.

For temperature and humidity control:

Humidity \ Temperature	Cool	Warm	Hot
Dry	Low	Low	Medium
Normal	Low	Medium	High
Humid	Medium	High	Very High

Each cell defines a fan-speed consequent.

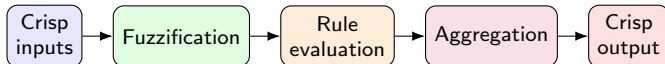
Two example rules

IF Temperature is Warm AND Humidity is Humid THEN Fan is High.

IF Temperature is Hot AND Humidity is Normal THEN Fan is High.

Fuzzy reasoning combines several partially active rules:

- 1 determine the membership of each input;
- 2 calculate each rule firing strength;
- 3 generate each rule's consequent;
- 4 combine the consequents;
- 5 produce a final decision.



“Fuzzy reasoning is the process of using fuzzy input memberships and fuzzy IF–THEN rules to infer an output, while allowing several rules to contribute simultaneously according to their firing strengths.”

1. Model

A structure that maps inputs to outputs.

ANN example:

$\mathbf{x} \rightarrow \text{Neural Network} \rightarrow \hat{y}$

- Network architecture
- Weights and biases
- Activation functions

Performs the mapping

2. Training

Adjusting model parameters using known data.

For an ANN:

$(\mathbf{x}, y) \rightarrow \text{Prediction} \rightarrow \text{Error}$

Error $\rightarrow$ BP $\rightarrow$ Update weights

Learning the parameters

3. Inference

Using the trained model on a new input.

Example:

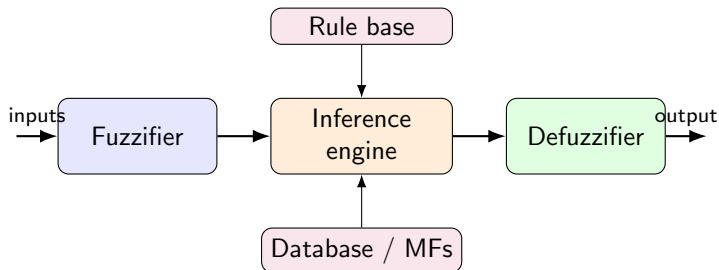
$\mathbf{x}_{\text{new}} = [T, V]$

$\mathbf{x}_{\text{new}} \rightarrow \text{Trained ANN} \rightarrow \hat{y}$

Using the learned model

Example

An ANN is the **model**. When we update its weights using training data, the process is called **training**. Once the trained model is deployed and used with new data, the process is called **inference**.



- Database stores membership-function definitions.
- Rule base stores expert IF–THEN knowledge.
- Inference engine determines how rules interact.

Inputs

Temperature: Cool, Warm, Hot
Humidity: Dry, Normal, Humid

Output

Fan command: Slow, Medium, Fast
Numerical range: 0–100%

Test measurement

$$T = 32^{\circ}\text{C}, \quad H = 70\%.$$

Assume

$$\mu_{\text{Warm}} = 0.8, \quad \mu_{\text{Hot}} = 0.2,$$

$$\mu_{\text{Normal}} = 0.3, \quad \mu_{\text{Humid}} = 0.7.$$

We will process the same measurement using Mamdani and Sugeno inference.

In a Mamdani system, both antecedents and consequents are fuzzy sets:

IF T is Hot AND H is Humid THEN Fan is Fast.

Typical processing:

- 1 Fuzzify the crisp inputs.
- 2 Apply fuzzy operators in each rule.
- 3 Clip or scale each output membership function.
- 4 Aggregate all output fuzzy sets.
- 5 Defuzzify the aggregated set.

Why engineers like it

The rule base is intuitive because the consequent remains linguistic and can be inspected visually.

Mamdani Example: Which Rules Fire?

For $T = 32^{\circ}\text{C}$ and $H = 70\%$:

Rule	Firing strength w_i
IF T is Warm AND H is Normal THEN Fan is Medium	$\min(0.8, 0.3) = 0.3$
IF T is Warm AND H is Humid THEN Fan is Fast	$\min(0.8, 0.7) = 0.7$
IF T is Hot AND H is Normal THEN Fan is Fast	$\min(0.2, 0.3) = 0.2$
IF T is Hot AND H is Humid THEN Fan is Fast	$\min(0.2, 0.7) = 0.2$

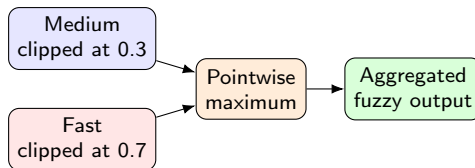
After max aggregation, *Medium* is activated to 0.3 and *Fast* to 0.7.

For rule i , minimum implication clips the consequent:

$$\mu_{B'_i}(y) = \min(w_i, \mu_{B_i}(y)).$$

All rule outputs are commonly combined using maximum aggregation:

$$\mu_B(y) = \max_i \mu_{B'_i}(y).$$



The centroid method calculates the centre of area:

$$y^* = \frac{\int y \mu_B(y) dy}{\int \mu_B(y) dy}.$$

For discrete output samples,

$$y^* = \frac{\sum_j y_j \mu_B(y_j)}{\sum_j \mu_B(y_j)}.$$

Other methods

Bisector of area, mean of maxima, smallest of maxima, and largest of maxima.

Defuzzification converts the combined linguistic recommendation into a command that an actuator can execute.

Suppose centroid defuzzification gives

$$y^* = 76\%.$$

The fan receives a 76% command rather than jumping between OFF and ON.

- Input transitions produce gradual output transitions.
- Several rules contribute to the decision.
- The reasoning can be explained using linguistic rules.

But remember

A fuzzy controller does not automatically guarantee stability or good performance. The rule base, membership functions, scaling, delays, and plant dynamics must still be validated.

In a Sugeno system, the consequent is a constant or function:

$$\text{IF } x \text{ is } A_i \text{ AND } y \text{ is } B_i \text{ THEN } z_i = p_i x + q_i y + r_i.$$

- Zero-order Sugeno: $z_i = c_i$.
- First-order Sugeno: $z_i = p_i x + q_i y + r_i$.
- Rule antecedents remain fuzzy.
- The final output is obtained by a weighted average.

Sugeno replaces an output fuzzy set with a compact mathematical consequent.

Use the same firing strengths, but assign constants:

$$z_{Medium} = 50, \quad z_{Fast} = 85.$$

The four active rules produce

$$(w_i, z_i) = (0.3, 50), (0.7, 85), (0.2, 85), (0.2, 85).$$

The weighted-average output is

$$z^* = \frac{\sum_i w_i z_i}{\sum_i w_i} = \frac{0.3(50) + 0.7(85) + 0.2(85) + 0.2(85)}{1.4} = 77.5\%.$$

No separate output aggregation curve or centroid calculation is required.

Consider two operating regions:

R_1 : IF Temperature is Warm THEN $z_1 = 1.2T + 0.2H + 10$,

R_2 : IF Temperature is Hot THEN $z_2 = 1.8T + 0.1H + 12$.

For normalized firing strengths $\bar{w}_i = w_i / (w_1 + w_2)$,

$$z^* = \bar{w}_1 z_1 + \bar{w}_2 z_2.$$

Interpretation

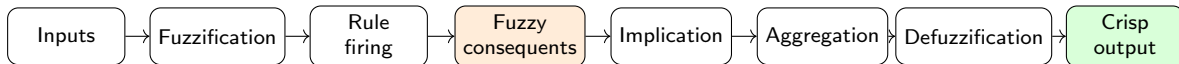
Each rule represents a local input–output model. Fuzzy weighting blends these models smoothly across operating regions.

Property	Mamdani	Sugeno
Consequent Interpretability	Fuzzy set Very intuitive linguistic output	Constant or function Antecedents intuitive; output mathematical
Computation	Aggregation and defuzzification required	Efficient weighted average
Optimization	Less convenient	Well suited to parameter learning
Typical use	Expert control and decision support	Adaptive control, modelling, ANFIS

Selection rule

Choose Mamdani when transparent linguistic reasoning is the priority; choose Sugeno when computational efficiency, optimization, or data-driven adaptation is important.

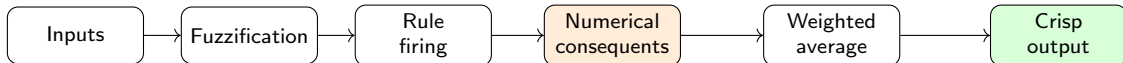
Mamdani FIS



Example:

Fan is Medium/Fast → clip → combine → centroid → 76%

Sugeno FIS



Example:

$$z_i = 50, 85 \rightarrow \frac{\sum w_i z_i}{\sum w_i} \rightarrow 77.5\%$$

Key difference

Both are similar up to **rule firing**. Mamdani uses **fuzzy consequents**, while Sugeno uses **numerical consequents**.

Inputs: human distance d and relative approach speed v . Output: cobot speed command u .

- IF Distance is Near THEN Speed is Stop.
- IF Distance is Medium AND Approach is Fast THEN Speed is Slow.
- IF Distance is Far AND Approach is Slow THEN Speed is Normal.

Why fuzzy reasoning helps

It provides smooth speed adaptation near overlapping operating regions and allows safety knowledge to be expressed clearly.

Safety warning

Fuzzy logic may support supervisory behaviour, but certified safety functions, limits, sensing architecture, and applicable robot-safety standards remain mandatory.

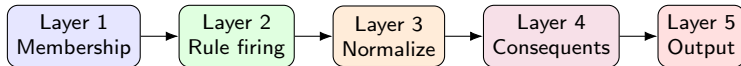
Our expert-designed controller works, but several questions remain:

- Are the membership-function centres correct?
- Should the curves be narrower or wider?
- Are the Sugeno coefficients optimal?
- Can experimental input–output data improve the model?

The engineer wants the interpretability of fuzzy rules and the parameter-learning ability of neural networks.

This leads to the Adaptive Neuro-Fuzzy Inference System (ANFIS).

ANFIS represents a Sugeno fuzzy inference system as a layered adaptive network.



- Premise parameters define membership functions.
- Consequent parameters define Sugeno output functions.
- Training adjusts these parameters to reduce prediction error.

Let the inputs be x and y :

$$R_1 : \text{IF } x \text{ is } A_1 \text{ AND } y \text{ is } B_1 \text{ THEN } f_1 = p_1x + q_1y + r_1,$$

$$R_2 : \text{IF } x \text{ is } A_2 \text{ AND } y \text{ is } B_2 \text{ THEN } f_2 = p_2x + q_2y + r_2.$$

The final output is

$$f = \frac{w_1 f_1 + w_2 f_2}{w_1 + w_2} = \bar{w}_1 f_1 + \bar{w}_2 f_2.$$

This computation is distributed across five network layers.

Each adaptive node calculates a membership degree:

$$O_{1,i} = \mu_{A_i}(x), \quad O_{1,j} = \mu_{B_j}(y).$$

For a generalized bell membership function,

$$\mu_A(x) = \frac{1}{1 + \left| \frac{x-c}{a} \right|^{2b}}.$$

Parameters a , b , and c determine width, slope, and centre. They are called **premise parameters** and are adjusted during training.

Layer 2 calculates firing strengths, commonly using product:

$$w_1 = \mu_{A_1}(x)\mu_{B_1}(y), \quad w_2 = \mu_{A_2}(x)\mu_{B_2}(y).$$

Layer 3 normalizes them:

$$\bar{w}_i = \frac{w_i}{w_1 + w_2}.$$

Therefore,

$$\bar{w}_1 + \bar{w}_2 = 1.$$

Meaning

The normalized firing strength specifies how strongly each local Sugeno model contributes to the current prediction.

Layer 4 calculates the weighted consequent:

$$O_{4,i} = \bar{w}_i f_i = \bar{w}_i (p_i x + q_i y + r_i).$$

Layer 5 sums the rule contributions:

$$O_{5,1} = \sum_i \bar{w}_i f_i = \frac{\sum_i w_i f_i}{\sum_i w_i}.$$

The parameters p_i , q_i , and r_i are **consequent parameters**.

ANFIS remains a fuzzy rule system, but its parameters are tuned from data.

For training samples $(\mathbf{x}_k, y_k)$, minimize an error such as

$$E = \frac{1}{N} \sum_{k=1}^N (y_k - \hat{y}_k)^2.$$

The common hybrid algorithm alternates:

- ➊ **Forward pass:** fix premise parameters and estimate consequent parameters using least squares.
- ➋ **Backward pass:** fix consequent parameters and update premise parameters using gradient descent.

Practical requirement

Use separate training, validation, and test data. Stop or regularize training if validation error begins to increase.

- ① Collect representative input–output data.
- ② Clean, normalize, and divide the data chronologically or by operating runs.
- ③ Select relevant input variables.
- ④ Initialize membership functions and rules using expert knowledge or clustering.
- ⑤ Train premise and consequent parameters.
- ⑥ Validate accuracy, smoothness, extrapolation, and physical plausibility.
- ⑦ Deploy with operating limits and monitoring.

Good data coverage is more important than simply increasing the number of rules.

Strengths

- Combines fuzzy interpretability with parameter learning.
- Models nonlinear relationships.
- Produces smooth input–output mappings.
- Can embed engineering knowledge at initialization.

Limitations

- Rule count can grow exponentially with inputs.
- Poor data produce poor rules and parameters.
- Interpretability decreases with many inputs or learned rules.
- Extrapolation outside training coverage is unreliable.

Rule explosion

With m membership functions for each of n inputs, grid partitioning may create m^n rules.

The factory now asks:

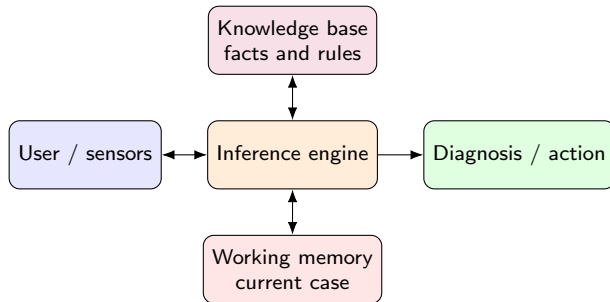
“Can the knowledge of our maintenance experts be made available during every shift?”

A rule-based expert system stores domain knowledge and applies it to a case.

Motor diagnosis

IF vibration is high AND temperature is high AND speed is normal, THEN suspect bearing degradation.

Unlike a controller, the system may return a diagnosis, explanation, confidence, or recommended action.



Additional components may include a knowledge-acquisition tool and an explanation facility that reports which rules produced the conclusion.

Forward chaining: data driven

Start with available facts, fire applicable rules, and continue until conclusions are reached.

facts $\rightarrow$ rules $\rightarrow$ conclusion

Useful for monitoring and event-driven diagnosis.

Backward chaining: goal driven

Start with a hypothesis and search for facts or rules that can confirm it.

hypothesis $\rightarrow$ required evidence

Useful for troubleshooting and consultation.

- ➊ Define the problem boundary and intended users.
- ➋ Acquire knowledge from experts, manuals, failure records, and experiments.
- ➌ Represent facts, rules, priorities, and uncertainty.
- ➍ Select forward, backward, or hybrid inference.
- ➎ Resolve conflicts when several rules can fire.
- ➏ Build an explanation facility.
- ➐ Verify rules and validate decisions using known and unseen cases.
- ➑ Maintain the knowledge base as equipment and procedures change.

Component	Role	Example
Fuzzy sets	Describe gradual conditions	Warm motor, high vibration
Fuzzy rules	Encode expert reasoning	IF hot AND high vibration THEN risk high
Mamdani FIS	Explainable risk assessment	Output: Low/Medium/Critical
Sugeno FIS	Efficient numerical estimate	Remaining-health score
ANFIS	Tune mapping from data	Learn from run-to-failure records
Expert system	Combine facts and actions	Recommend inspection and explain why

These methods are complementary parts of an intelligent mechatronic decision system.

Design a fuzzy supervisor for an autonomous mobile robot.

Given

Inputs: obstacle distance and battery level.

Output: commanded speed.

In groups, define:

- 1 three linguistic terms for each input and output;
- 2 suitable membership-function shapes and ranges;
- 3 at least five IF–THEN rules;
- 4 one potentially unsafe rule conflict;
- 5 whether Mamdani or Sugeno is more suitable, with justification.

- 1 Can an input belong to two fuzzy sets simultaneously?
- 2 Is a membership degree the same as probability?
- 3 What is the role of defuzzification?
- 4 How does a Sugeno consequent differ from a Mamdani consequent?
- 5 Which ANFIS parameters define membership functions?
- 6 When would forward chaining be preferred?

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- ❹ How does a Sugeno consequent differ from a Mamdani consequent?
- ❺ Which ANFIS parameters define membership functions?
- ❻ When would forward chaining be preferred?

Expected ideas

Yes; no; convert fuzzy output to a crisp value; fuzzy set versus function/constant; premise parameters; when facts arrive and conclusions must be generated.

- Fuzzy sets represent gradual membership using values in $[0, 1]$.
- Membership functions translate linguistic concepts into mathematics.
- Fuzzy rules combine expert knowledge with sensor measurements.
- Mamdani inference uses fuzzy consequents and explicit defuzzification.
- Sugeno inference uses constant or functional consequents.
- ANFIS tunes a Sugeno fuzzy system using data.
- Rule-based expert systems organize knowledge, inference, explanation, and action.

Fuzzy logic is most valuable when precise measurements must be combined with approximate but useful human expertise.

Instructions

- Scan the QR code to access the quiz.
- Sign in using your **VIT Google account**.
- The quiz contains **10 questions**.
- Select the most appropriate answer.
- Only **one submission per account** is allowed.

QUIZ: Scan to Begin



Fuzzy Logic Converts Human Expertise into Intelligent Decisions

Thank You